

Double Lower Limb Kinematics-based Invariant Extended Kalman Filtering for Pedestrian Inertial State Estimation *

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Abstract—This paper combined the double lower limbs and waist as rigid bodies. Each angle of hip, knee, and ankle joint can be measured through several Inertial Measurement Unit (IMU), formulating the double lower kinematics. Focusing on the model uncertainty of stance and swing phase switching within gait cycle, a novel invariant extended Kalman filter (EKF) was contrived to produce state estimation. Towards the rotation path, the people executed walking locomotion in simulation. Compared to the classical EKF, the more rapid error convergence for pedestrian state estimation is accomplished using the invariant extended Kalman filter (InEKF). This methodology demonstrated the validity, feasibility and accuracy of pedestrian inertial state estimation with InEKF, which expanded the application scope of inertial measurement apparatus.

Keywords—Invariant Kalman filter, lower limb kinematics, state estimation, Inertial Measurement Unit

I. INTRODUCTION

With the continuous increase in people's demand for precise position and motion state, the widespread employ of portable device promotes the development of localization system. The technology of motion state acquirement has applied in several aspects of life, including the human-machine interaction[1], motion capture[2], and exoskeleton[3]. The IMU-based pedestrian state estimation is a kind of micro-electromechanical system, whose benefits are high autonomy, intellectuality, reliability and low cost.

On account of the existence of stance phase that foot can maintain relative stillstand with the ground. The Zero Velocity Update (ZUPT) solved the error accumulation issues[4]. During the Zero-Velocity period, Kalman Filter is usually used to correct systematic errors. Specifically, importing the observed quantity is to construct constrain and correct error[5][6]. To restrain the error accumulation resulted from integral operator, the Pedestrian Dead Reckoning[7] computes the real-time position of pedestrian through monitoring the locomotion data containing the step length, step frequency, and orientation. The common approaches of step length works through fitting acceleration including the linear and nonlinear model. However, the step size estimation determines the parameters through collecting gait features. So the accuracy of step length model relies on the gait statistical information.

The discrete and continuous states are scheduled in pedestrian locomotion. Apart from the stance phase, a typical

gait involves the swing phase which forms variable-speed trajectory. The selection of sensor position can influence the measured acceleration variation when foot contacts with ground, leading to the low zero-speed detection accuracy[8]. Placing the IMU on feet, the instantaneous impact generated from touching the ground[9] causes extreme value in acceleration measurement. The faster the walking speed is, the shorter the relative stationary time of ground touching will be. It becomes apparent that the error of inertial system grows bigger. Whereas, while putting IMU on the lower leg and waist, the occurrence of sensor data abnormality originated from strenuous exercise decreases due to the lower impulse.

Adding sensors[10] can increase the observed value to boost the positional accuracy of system. But utilizing the visual sensors to assistant localization enhances system cost[11]. Even though novel sensor arrangement can boost the localization accuracy, but also it may reduce the system portability and increase complexity.

These days the disparate Kalman filters[12] have been used to estimate body pose and velocity through the nonlinear forward kinematics[13]. The EKF is also applied to additionally process the inaccuracy of the nonlinear kinematics chain and sensor placement imperfections. Nevertheless, the EKF has pretty high requirements[14] for the accuracy of initial values. If the difference between initial value setup of state and practical situation emerges pretty big, the filter is hard to converge. Furthermore, the EKF can leads to inconsistency issues[15]. The EKF calculate current covariance matrix through linearization of previous state, while the actual value of current state covariance matrix does not stay the same with it. The InEKF applies group transformation to state variables based on Lie algebra and changes the defined way of errors. The independence of the system matrix and the state estimated values are realized, expanding the state space that ensures the convergency of state and covariance[16].

This work used the pedestrian waist and double lower limbs, which are considered as rigid bodies under the condition without bringing new type sensor. Different with step length estimation based on empirically fitted model such as pedestrian dead reckoning (PDR)[17] and zero velocity update(ZUPT)[18], this work placed IMU on waist, thighs and calves of two legs, as well as feet. Fashion the low-limbs kinematics to estimate the pedestrian motion state. Fusing IMU data with InEKF restrained estimation error, furtherly improving the accuracy.

II. INVARIANT EXTENDED KALMAN FILTERING

This section involved some fundamental concepts related to the special orthogonal group $SO(3)$ and its associated Lie algebra $so(3)$, which are essential to speculate the InEKF theory.

The elements in special orthogonal group $SO(3)$ consist of 3×3 real matrix R that satisfy:

$$R^T R = I \text{ and } \det(R) = 1$$

where I is the 3×3 identity matrix. $SO(3)$ is a matrix Lie group, and the associated Lie algebra is written as $so(3)$, which is a 3×3 skew-symmetric matrix. For the convenience of calculation, the linear mapping from R^3 to $so(3)$ is defined as:

$$\mathbf{w}^\wedge = \begin{bmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{bmatrix} \quad (1)$$

where $w = [w_1, w_2, w_3]^T \in \mathbb{R}^3$.

The connection between $so(3)$ and $SO(3)$ is constructed from exponential map exp:

$$\exp(\mathbf{w}^\wedge) = \sum_{n=0}^{\infty} \frac{(\mathbf{w}^\wedge)^n}{n!} = I + \frac{\sin\|\mathbf{w}\|}{\|\mathbf{w}\|} \mathbf{w}^\wedge + \frac{1-\cos\|\mathbf{w}\|}{\|\mathbf{w}\|^2} (\mathbf{w}^\wedge)^2$$

in which $\|\cdot\|$ denotes the standard vector norm. For simplicity, we denote $\exp(\mathbf{w}^\wedge) = \text{Exp}(\mathbf{w})$.

For any $R \in SO(3)$ and $\mathbf{w}^\wedge \in so(3)$, the adjoint mapping is defined as:

$$R \mathbf{w}^\wedge R^T = (R \mathbf{w})^\wedge \quad (2)$$

Additionally, the left Jacobian $J(\mathbf{w}^\wedge)$ of $SO(3)$ is defined as:

$$J(\mathbf{w}^\wedge) = \sum_{n=0}^{\infty} \frac{(\mathbf{w}^\wedge)^n}{(n+1)!} \quad (3)$$

which can be expressed as:

$$J(\mathbf{w}^\wedge) = \frac{\sin\|\mathbf{w}\|}{\|\mathbf{w}\|} I + \left(1 - \frac{\sin\|\mathbf{w}\|}{\|\mathbf{w}\|}\right) \mathbf{a} \mathbf{a}^T + \frac{1 - \cos\|\mathbf{w}\|}{\|\mathbf{w}\|} \mathbf{a}^\wedge$$

where $\|\mathbf{w}\|$ is the rotation angle and $\mathbf{a} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$ is the rotation axis with unit length.

For a continuous trajectory $R(t)$ in motion on $SO(3)$, the dynamics are given by:

$$\frac{d}{dt} R = f_w(R) = R \mathbf{w}^\wedge = R \mathbf{w}^\wedge R^T R = (R \mathbf{w})^\wedge R \quad (4)$$

For the trajectory $R(t)$ and its estimation $\hat{R}(t)$ in $SO(3)$, there are two methods to define the error between them:

- Right-invariant error: $\boldsymbol{\eta} = \hat{R} R^T$
- Left-invariant error: $\boldsymbol{\eta} = R^T \hat{R}$

The above-mentioned error remains constant for any group element in $X \in SO(3)$. But the common linear error $R - \hat{R}$ is not an element in $SO(3)$. The right-invariant error is used in this paper due to its convenience in calculation. The dynamics $f_w(R)$ satisfies the "group affine" property:

$$\begin{aligned} f_w(R_1 R_2) &= R_1 R_2 \mathbf{w}^\wedge \\ &= f_w(R_1) R_2 + R_1 f_w(R_2) - R_1 f_w(I) R_2 \end{aligned}$$

This property ensures that the error dynamics is trajectory-independent. That is

$$\begin{aligned} \frac{d}{dt} \boldsymbol{\eta}^r &= \frac{d}{dt} (\hat{R} R^T) = \frac{d}{dt} (\hat{R}) R^T + \hat{R} \frac{d}{dt} (R^T) \\ &= \hat{R} \mathbf{w}^\wedge R^T + \hat{R} (R \mathbf{w}^\wedge)^T = \hat{R} \mathbf{w}^\wedge R^T - \hat{R} \mathbf{w}^\wedge R^T \\ &= 0 \\ \frac{d}{dt} \boldsymbol{\eta}^l &= \frac{d}{dt} (R^T \hat{R}) = \frac{d}{dt} (R^T) \hat{R} + R^T \frac{d}{dt} (\hat{R}) \\ &= -\mathbf{w}^\wedge R^T \hat{R} + R^T \hat{R} \mathbf{w}^\wedge = \boldsymbol{\eta}^l \mathbf{w}^\wedge - \mathbf{w}^\wedge \boldsymbol{\eta}^l = 0 \end{aligned}$$

For stochastic variables in $SO(3)$, the state is represented as:

$$R = \exp(\boldsymbol{\varepsilon}^\wedge) \hat{R} \quad (5)$$

where $\hat{R} \in SO(3)$ is the prescribed, and $\boldsymbol{\varepsilon} \in \mathbb{R}^3$ is a zero-mean Gaussian process with covariance $\mathbf{P}_\varepsilon$, that is $\boldsymbol{\varepsilon} \sim N(0, \mathbf{P}_\varepsilon)$. The R is referred as right-translated exponential noise with right-invariant covariance $\mathbf{P}_\varepsilon$, denoted as $R \sim N_R(\hat{R}, \mathbf{P}_\varepsilon)$.

III. SYSTEM MODEL

To estimate the orientation, velocity, and position of the body (IMU) in the world frame, an auxiliary inertial navigation is enacted. The motion states are described as $\mathbf{R}_{WB}(t)$, $\mathbf{v}_{WB}(t)$, and $\mathbf{p}_{WB}(t)$ separately. Additionally, attach the foot contact points $\mathbf{p}_{wc}(t)$ in the world frame. The above-mentioned state variables formulates a matrix Lie Group $\mathcal{G}$. During the period of N contact points, the state $\mathbf{X}_t \in SE_{N+2}(3)$ is defined by the following matrix.

$$\mathbf{X}_t \triangleq \begin{bmatrix} \mathbf{R}_t & \mathbf{v}_t & \mathbf{p}_t & \mathbf{d}_t \\ \mathbf{0}_{1,3} & 1 & 0 & 0 \\ \mathbf{0}_{1,3} & 0 & 1 & 0 \\ \mathbf{0}_{1,3} & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{u}_t = \begin{bmatrix} \tilde{\boldsymbol{\omega}}_{WB}(t) \\ \tilde{\mathbf{a}}_{WB}(t) \end{bmatrix} \triangleq \begin{bmatrix} \tilde{\boldsymbol{\omega}}_t \\ \tilde{\mathbf{a}}_t \end{bmatrix}$$

in which the input $\mathbf{u}_t$ is arranged from the angular velocity $\tilde{\boldsymbol{\omega}}_{WB}(t)$ and linear acceleration $\tilde{\mathbf{a}}_{WB}(t)$ measuring from the IMU in the body (or IMU) frame. The Lie algebra of $\mathcal{G}$ is an $N + 5$ dimensional square matrix, which is expressed by $\mathbf{g}$. It is important to establish a map $\mathcal{L}_q: \mathbb{R}^{3N+9}$ [19] that maps a vector to the corresponding element of the Lie algebra. When people contacts single foot with ground, for instance this function is characterized as:

$$\mathcal{L}_q(\boldsymbol{\xi}) = \begin{bmatrix} (\boldsymbol{\xi}^R)_\times & \boldsymbol{\xi}^v & \boldsymbol{\xi}^p & \boldsymbol{\xi}^d \\ \mathbf{0}_{1,3} & 0 & 0 & 0 \\ \mathbf{0}_{1,3} & 0 & 0 & 0 \\ \mathbf{0}_{1,3} & 0 & 0 & 0 \end{bmatrix}, \quad (6)$$

where $(\cdot)_\times$ represents a 3×3 skew-symmetric matrix. The adjoint operator is shown as follows.

$$\text{Ad}_{\mathbf{x}_t} = \begin{bmatrix} \mathbf{R} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (\mathbf{v}_t)_\times \mathbf{R}_t & \mathbf{R}_t & \mathbf{0} & \mathbf{0} \\ (\mathbf{p}_t)_\times \mathbf{R}_t & \mathbf{0} & \mathbf{R}_t & \mathbf{0} \\ (\mathbf{d}_t)_\times \mathbf{R}_t & \mathbf{0} & \mathbf{0} & \mathbf{R}_t \end{bmatrix}. \quad (7)$$

IV. RIGHT-INVARIANT EKF FOR DOUBLE LOWER KINEMATICS-BASED KALMAN MODEL

In practical realization, the measurements from IMU sensors always contains the gyroscope and accelerometer biases, desiring additional modeling. There are two approaches to handle these appended states. The one is to set the biases as constant parameters. Assume that they usually change over a longer period of time than the experiment[20]. Then we can calibrate these biases in a separate experiment. Another one method regards the biases as a part of the estimated states. Then the deviation can be estimated simultaneously while estimating the attitude. The latter scheme is adopted in this paper, because the on-line real-time compensation of the influence of deviation can improve the accuracy of attitude estimation. Once the deviation is taken into account, no Lie group can satisfy the group affine properties of the dynamic equation[21]. Even though several theoretical characters are no longer valid, it is still possible to design an “augmented InEKF” that still outperforms the “standard EKF”[22]. Due to that the two fashions of processing deviation are similar, we only discuss gyroscope bias in this section for the purpose of saving text space.

A. Containing IMU Biases

The gyroscope bias is slow-varying signal which disturbs the measured results in an additive way.

$$\tilde{\boldsymbol{\omega}}_t = \boldsymbol{\omega}_t + \mathbf{b}_t^g + \mathbf{w}_t^g, \quad \mathbf{w}_t^g \sim \mathcal{N}(\mathbf{0}_{3,1}, \Sigma^g) \quad (8)$$

The gyroscope bias builds a parameter vector requiring accurate estimation to be a part of the RI-EKF states.

$$\boldsymbol{\theta}_t \triangleq \mathbf{b}^g(t) \triangleq \mathbf{b}_t^g \in \mathbb{R}^6 \quad (9)$$

The state of model becomes the original matrix Lie group and bias vector $(\mathbf{X}_t, \boldsymbol{\theta}_t) \in \mathcal{G} \times \mathbb{R}^6$. Now define the right-invariant error as follows:

$$\mathbf{e}_t^r \triangleq (\tilde{\mathbf{X}}_t \mathbf{X}_t^{-1}, \tilde{\boldsymbol{\theta}}_t - \boldsymbol{\theta}_t) \triangleq (\boldsymbol{\eta}_t^r, \boldsymbol{\zeta}_t) \quad (10)$$

in which the right-invariant error is

$$\boldsymbol{\eta}_t^r = \begin{bmatrix} \hat{\mathbf{R}}_t \mathbf{R}_t^T & \hat{\mathbf{v}}_t - \hat{\mathbf{R}}_t \mathbf{R}_t^T \mathbf{v}_t & \hat{\mathbf{p}}_t - \hat{\mathbf{R}}_t \mathbf{R}_t^T \mathbf{p}_t & \hat{\mathbf{d}}_t - \hat{\mathbf{R}}_t \mathbf{R}_t^T \mathbf{d}_t \\ \mathbf{0}_{1,3} & 1 & 0 & 0 \\ \mathbf{0}_{1,3} & 0 & 1 & 0 \\ \mathbf{0}_{1,3} & 0 & 0 & 1 \end{bmatrix}$$

and the bias vector can be written as

$$\boldsymbol{\zeta}_t = \hat{\mathbf{b}}_t^g - \mathbf{b}_t^g \triangleq \boldsymbol{\zeta}_t^g \quad (11)$$

Now the deterministic system dynamics depends on the system inputs $\mathbf{u}_t$ and the bias parameters $\boldsymbol{\theta}_t$:

$$f(\boldsymbol{\theta}_t, \mathbf{u}_t)(\mathbf{X}_t) = \begin{bmatrix} \mathbf{R}_t(\tilde{\boldsymbol{\omega}}_t - \mathbf{b}_t^g)_\times & \mathbf{R}_t(\tilde{\boldsymbol{\alpha}}_t - \mathbf{b}_t^a) + \mathbf{g} & \mathbf{v}_t & \mathbf{0}_{3,1} \\ \mathbf{0}_{1,3} & 1 & 0 & 0 \\ \mathbf{0}_{1,3} & 0 & 1 & 0 \\ \mathbf{0}_{1,3} & 0 & 0 & 1 \end{bmatrix}$$

The bias $\mathbf{b}_t^g$ dynamically employs the typical “Brown Motion” model to model. Therefore, its derivative is white gaussian noise, capturing the slow-time-varying property of parameter:

$$\dot{\mathbf{b}}_t^g = \mathbf{w}_t^{bg} \quad (12)$$

For the purpose of calculating the linearizing error dynamic state, firstly differentiate the augmented right-invariant error.

$$\frac{d}{dt} \mathbf{e}_t^r = \left(\frac{d}{dt} \boldsymbol{\eta}_t^r, \mathbf{w}_t^{bg} \right). \quad (13)$$

After executing the chain rule and performing a first order approximation, $\boldsymbol{\eta}_t^r = \exp(\boldsymbol{\xi}_t) \approx \mathbf{I}_a + \mathcal{L}_g(\boldsymbol{\xi}_t)$, the augmented invariant error dynamics estimates the trajectory only depending on the noise and bias error $\boldsymbol{\zeta}_t$. When there is no noise and bias, the error dynamics has no dependence on estimating trajectory. Now construct a linear system:

$$\frac{d}{dt} \begin{bmatrix} \boldsymbol{\xi}_t \\ \boldsymbol{\zeta}_t \end{bmatrix} = \mathbf{A}_t \begin{bmatrix} \boldsymbol{\xi}_t \\ \boldsymbol{\zeta}_t \end{bmatrix} + \begin{bmatrix} \text{Ad}_{\tilde{\mathbf{x}}_t} & \mathbf{0}_{12,6} \\ \mathbf{0}_{6,12} & \mathbf{I}_6 \end{bmatrix} \mathbf{w}_t \quad (14)$$

where the block structure $\mathbf{A}_t = \begin{bmatrix} \mathbf{A}_x & \mathbf{A}_{x,\theta} \\ \mathbf{0} & \mathbf{A}_\theta \end{bmatrix}$. And the noise vector is outlined as $\mathbf{w}_t \triangleq \text{vec}(\mathbf{w}_t^g, \mathbf{0}_{1,3}, \mathbf{h}_R(\tilde{\boldsymbol{\alpha}}_t) \mathbf{w}_t^v, \mathbf{w}_t^{bg})$.

B. Right-invariant Forward Kinematic Measurement Model

The $\tilde{\boldsymbol{\alpha}}_t \in \mathbb{R}^M$ indicates the joint position vector between the body and the contact points. It is assumed that the encoder measurement is additionally disrupted by the white Gaussian noise.

$$\tilde{\boldsymbol{\alpha}}_t = \boldsymbol{\alpha}_t + \mathbf{w}_t^\alpha, \quad \mathbf{w}_t^\alpha \sim \mathcal{N}(\mathbf{0}_{M,1}, \Sigma^\alpha) \quad (15)$$

Combining the forward kinematics, define the contact point position in the body frame in the following equation.

$$\mathbf{p}_{BC}(t) \triangleq \mathbf{h}_p(\tilde{\boldsymbol{\alpha}}_t - \mathbf{w}_t^\alpha) \approx \mathbf{h}_p(\tilde{\boldsymbol{\alpha}}_t) - \mathbf{J}_v(\tilde{\boldsymbol{\alpha}}_t) \mathbf{w}_t^\alpha \quad (16)$$

in which the Jacobian matrix $\mathbf{J}_v$ corresponds to linear velocity[15]. Therefore, the position measurement result of forward kinematics yields

$$\mathbf{h}_p(\tilde{\boldsymbol{\alpha}}_t) = \mathbf{R}_t^T(\mathbf{d}_t - \mathbf{p}_t) + \mathbf{J}_v(\tilde{\boldsymbol{\alpha}}_t) \mathbf{w}_t^\alpha \quad (17)$$

It does not rely on the biases of IMU. As for the augmented right-invariant error, the matrix $\mathbf{H}_t$ can simply add zero to

satisfy the dimensionality requirements. The linear update equation becomes

$$\begin{bmatrix} \xi_t^+ \\ \zeta_t^+ \end{bmatrix} = \begin{bmatrix} \xi_t \\ \zeta_t \end{bmatrix} - \begin{bmatrix} \mathbf{K}_t^\xi \\ \mathbf{K}_t^\zeta \end{bmatrix} \left(\mathbf{H}_t \begin{bmatrix} \xi_t \\ \zeta_t \end{bmatrix} - \hat{\mathbf{R}}_t (\mathbf{J}_v(\tilde{\alpha}_t) \mathbf{w}_t^\alpha) \right) \quad (18)$$

And the matrix of measurement model is

$$\mathbf{H}_t = [\mathbf{H}_x \quad \mathbf{H}_\theta] \quad (19)$$

C. Continuous RI-EKF System

Finally, the augmented InEKF equation including gyroscope deviation can be integrated as follows. The estimated states utilizes the next differential equation to predict the propagation.

$$\left(\frac{d}{dt} \hat{\mathbf{x}}_t, \frac{d}{dt} \hat{\theta}_t \right) = \left(f(\hat{\theta}_t, \mathbf{u}_t)(\hat{\mathbf{x}}_t), \mathbf{0}_{6,1} \right) \quad (20)$$

The dynamic covariance of right-invariance error is calculated through solving the Riccati equation.

$$\frac{d}{dt} \mathbf{P}_t = \mathbf{A}_t \mathbf{P}_t + \mathbf{P}_t \mathbf{A}_t^T + \hat{\mathbf{Q}}_t \quad (21)$$

where the matrices $\mathbf{A}_t$ and $\hat{\mathbf{Q}}_t$ are explicit as follows:

$$\mathbf{A}_t = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\hat{\mathbf{R}}_t & \mathbf{0} \\ (\mathbf{g})_\times & \mathbf{0} & \mathbf{0} & \mathbf{0} & -(\hat{\mathbf{v}}_t)_\times \hat{\mathbf{R}}_t & -\hat{\mathbf{R}}_t \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & -(\hat{\mathbf{p}}_t)_\times \hat{\mathbf{R}}_t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -(\hat{\mathbf{d}}_t)_\times \hat{\mathbf{R}}_t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

$$\hat{\mathbf{Q}}_t = \begin{bmatrix} \text{Ad}_{\hat{\mathbf{x}}_t} & \mathbf{0}_{12,6} \\ \mathbf{0}_{6,12} & \mathbf{I}_6 \end{bmatrix} \text{Cov}(\mathbf{w}_t) \begin{bmatrix} \text{Ad}_{\hat{\mathbf{x}}_t} & \mathbf{0}_{12,6} \\ \mathbf{0}_{6,12} & \mathbf{I}_6 \end{bmatrix}^T$$

The update equation of state estimation is

$$(\hat{\mathbf{x}}_t^+, \hat{\theta}_t^+) = \left(\exp \left(\mathbf{K}_t^\xi \Pi(\hat{\mathbf{x}}_t \mathbf{Y}_t) \right) \hat{\mathbf{x}}_t, \hat{\theta}_t + \mathbf{K}_t^\xi \Pi(\hat{\mathbf{x}}_t \mathbf{Y}_t) \right)$$

in which the computing approaches of gains $\mathbf{K}_t^\xi$ and $\mathbf{K}_t^\zeta$ are computing are

$$\mathbf{S}_t = \mathbf{H}_t \mathbf{P}_t \mathbf{H}_t^T + \hat{\mathbf{N}}_t \quad (22)$$

$$\mathbf{K}_t = \mathbf{P}_t \mathbf{H}_t^T \mathbf{S}_t^{-1} \quad (23)$$

with the following measurement, output, and noise matrices,

$$\mathbf{H}_t^T = [\mathbf{h}_p^T(\tilde{\alpha}_t) \quad \mathbf{0} \quad \mathbf{1} \quad -\mathbf{1}]$$

$$\mathbf{H}_t = [\mathbf{0} \quad \mathbf{0} \quad -\mathbf{I} \quad \mathbf{I} \quad \mathbf{0} \quad \mathbf{0}]$$

$$\hat{\mathbf{N}}_t = \hat{\mathbf{R}}_t \mathbf{J}_v(\tilde{\alpha}_t) \text{Cov}(\mathbf{w}_t^\alpha) \mathbf{J}_v^T(\tilde{\alpha}_t) \hat{\mathbf{R}}_t^T$$

D. Discretization

In the previous derivation course, the filter state propagation is continuous-time. However, these equations need to be discrete

to achieve the filter process. For this purpose, we make the zero-order hold and operate Euler integration on IMU measurement inputs from t_k to t_{k+1} . The error dynamic is simple Gaussian noise, the discrete-form deterministic dynamic equation is

$$\hat{\mathbf{d}}_{k+1} = \hat{\mathbf{d}}_k, \quad \hat{\mathbf{b}}_{k+1}^g = \hat{\mathbf{b}}_k^g, \quad \hat{\mathbf{b}}_{k+1}^a = \hat{\mathbf{b}}_k^a \quad (24)$$

It is assumed that we can update the $\hat{\mathbf{R}}_{k+1}$ through exponential map in $SO(3)$ while performing zero-order hold on measured angular velocity.

$$\hat{\mathbf{R}}_{k+1} = \hat{\mathbf{R}}_k \exp \left((\hat{\omega}_k - \hat{\mathbf{b}}_k^g) \Delta t \right) \quad (25)$$

$$\hat{\mathbf{v}}_{k+1} = \hat{\mathbf{v}}_k + \hat{\mathbf{R}}_k (\tilde{\mathbf{a}}_k - \hat{\mathbf{b}}_k^a) \Delta t + \mathbf{g} \Delta t \quad (26)$$

$$\hat{\mathbf{p}}_{k+1} = \hat{\mathbf{p}}_k + \hat{\mathbf{v}}_k \Delta t + \frac{1}{2} \hat{\mathbf{R}}_k (\tilde{\mathbf{a}}_k - \hat{\mathbf{b}}_k^a) \Delta t^2 + \frac{1}{2} \mathbf{g} \Delta t^2 \quad (27)$$

where $\Delta t = t_{k+1} - t_k$. In order to update the covariance, we need to resolve the Riccati equation of continuous time, whose analytic solution is

$$\mathbf{P}_{k+1} = \Phi(t_{k+1}, t_k) \mathbf{P}_k \Phi(t_{k+1}, t_k)^T + \hat{\mathbf{Q}}_k \quad (28)$$

in which the discrete noise covariance matrix becomes

$$\hat{\mathbf{Q}}_k = \int_{t_k}^{t_{k+1}} \Phi(t_{k+1}, t_k) \hat{\mathbf{Q}}_t \Phi(t_{k+1}, t_k) dt \quad (29)$$

the state-transition matrix $\Phi(t_{k+1}, t_k)$ satisfies

$$\frac{d}{dt} \Phi(t, t_k) = \mathbf{A}_t \Phi(t, t_k), \quad \Phi(t_k, t_k) = \mathbf{I} \quad (30)$$

Solve the formula, the discrete-time linear model is organized by $\Phi(t_{k+1}, t_k) \triangleq (\mathbf{A}_t \Delta t)$. Similar to the state transition matrix, the discrete noise covariance matrix also has an analytical solution. Nevertheless, this matrix is usually approximated as $\hat{\mathbf{Q}}_k \approx \Phi(t_{k+1}, t_k) \hat{\mathbf{Q}}_t \Phi(t_{k+1}, t_k) \Delta t$ in practical application.

V. TRANSITING CONTACT AND SWING STATE

Part III and IV narrated the calculation flow of the RI-EKF for pedestrian locomotion. It is effective when touching point is constant with time-varying. Even though contacts with ground on pedestrian feet are discrete actions and determined as established and damaged. When pedestrian navigates through the surroundings, we found that it is critical to conveniently apply or remove the contact state in and from the observer.

A. Eliminating contact points

To removing the a previous contact point from the state, we simply eliminates the corresponding column or row from matrix Lie group to marginalize the variable. Meantime, the corresponding elements of the covariance matrix are also erased. This procedure can be accomplished through a simple linear transformation. For instance, when single contact becomes zeros contact in pedestrian walking, the newly reduced covariance can be speculated by the following formula.

$$\begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \end{bmatrix} \triangleq \mathbf{F}_r \xi_t \quad (31)$$

That is $\xi_t^{new} = \mathbf{F}_r \mathbf{P}_t \mathbf{F}_r^T$.

B. Applying Contact Points

While pedestrian is locomoting on the ground, the new contacts with environment are generating. The states and covariance matrix need to be augmented. Especially, we demands to pay attention to the mean and covariance with initializing the lately estimated contact points. If pedestrian turns into one contact from no contact as an example, the initial estimation can be attained from the following forward kinematics:

$$\hat{\mathbf{d}}_t = \hat{\mathbf{p}}_t + \hat{\mathbf{R}}_t \mathbf{h}_p(\tilde{\alpha}_t) \quad (32)$$

For the purpose of calculating the covariance, we focus on the right-invariant error.

$$\hat{\mathbf{d}}_t - \hat{\mathbf{R}}_t \mathbf{R}_t^T \mathbf{d}_t = \hat{\mathbf{p}}_t - \hat{\mathbf{R}}_t \mathbf{R}_t^T \mathbf{d}_t + \hat{\mathbf{R}}_t \mathbf{h}_p(\tilde{\alpha}_t) \quad (33)$$

$$\boldsymbol{\eta}_t^d = \hat{\mathbf{p}}_t - \hat{\mathbf{R}}_t \mathbf{R}_t^T (\mathbf{p}_t + \mathbf{R}_t \mathbf{h}_p(\tilde{\alpha}_t - \mathbf{w}_t^\alpha)) + \hat{\mathbf{R}}_t \mathbf{h}_p(\tilde{\alpha}_t)$$

$$\boldsymbol{\eta}_t^d \approx \boldsymbol{\eta}_t^p + \hat{\mathbf{R}}_t \mathbf{J}_v(\tilde{\alpha}_t) \mathbf{w}_t^\alpha \quad (34)$$

Thus, we have $\boldsymbol{\eta}_t^d = \boldsymbol{\eta}_t^p + \hat{\mathbf{R}}_t \mathbf{J}_v(\tilde{\alpha}_t) \mathbf{w}_t^\alpha$.

Accordingly, the covariance can be augmented using the below linear map.

$$\begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \xi_t^R \\ \xi_t^v \\ \xi_t^p \\ \xi_t^d \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \hat{\mathbf{R}}_t \mathbf{J}_v(\tilde{\alpha}_t) \end{bmatrix} \mathbf{w}_t^\alpha \quad (35)$$

$$\xi_t^{new} \triangleq \mathbf{F}_a \xi_t + \mathbf{G}_t \mathbf{w}_t^\alpha \quad (36)$$

$$\Rightarrow \mathbf{P}_t^{new} = \mathbf{F}_a \mathbf{P}_t \mathbf{F}_a^T + \mathbf{G}_t \text{Cov}(\mathbf{w}_t^\alpha) \mathbf{G}_t^T \quad (37)$$

VI. SIMULATION

The IMU utilized to measure the 10 freedoms of lower joints, is equipped on thighs, calves, and feet of double lower limbs. Combining the IMU fixed on waist to provide angular velocity and linear acceleration measurements, the kinematics of pedestrian locomotion is computed.

The EKF trial proceeded the same specification with InEKF run. The proposed algorithm and EKF flow were operated in MATLAB. The computation time of single filter loop for both algorithms was approximately 1ms, possessing low calculation consumption. The associated parameter initialization and specification are listed in Table I.

TABLE I. SIMULATION NOISE STATISTICS AND INITIAL COVARIANCE

Measurement Type	Noise st. dev.	State Element	Initial st. dev.
Linear Acceleration	0.04 m/s ²	Orientation of IMU	±5.0 deg
Angular Velocity	0.01 rad/s	Velocity of IMU	±0.05 m/s
Accelerometer Bias	0.01 m/s ²	Position of IMU	±5 m
Gyroscope Bias	0.001 rad/s	Position of Foot	±1m
Contact Linear Velocity	0.01 m/s	Gyroscope Bias	0.01 rad/s
		Accelerometer Bias	0.01 m/s

The body position, velocity, and posture along x, y direction and yaw estimates for both filters are shown in Fig. 1, 2, and 3. The nominated InEKF converges faster than EKF in all 60 runs, where the dashed black thick lines represented the practical motion states. Other colorful line indicated the estimation under different initial sample values. The EKF funs almost did not get also the stability and convergence with true values.

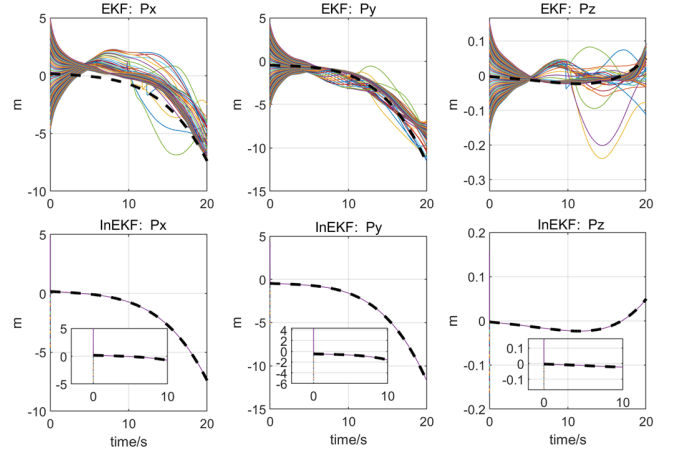


Fig.1 Position estimation comparison

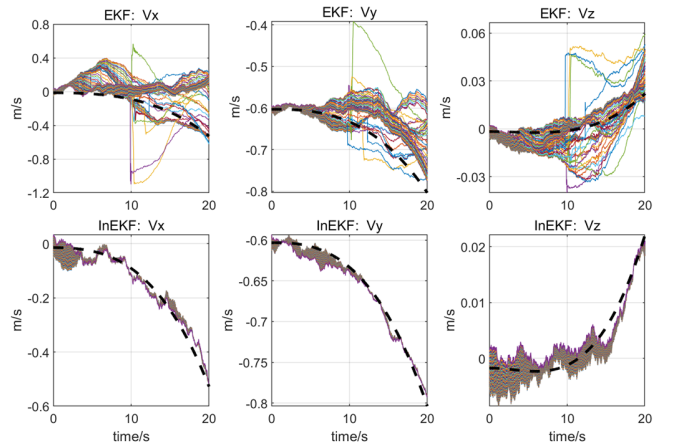


Fig.2 Velocity estimation comparison

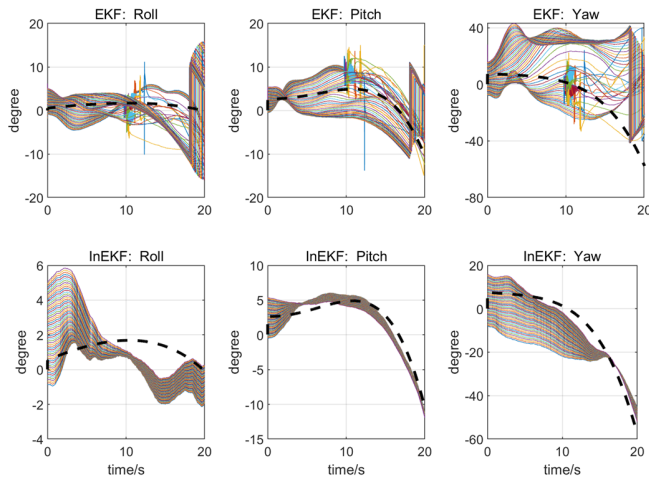


Fig.3 Posture estimation comparison

Compared to the typical EKF, the improved filter convergency rate attributed to the higher accuracy of the proposed estimation flow with large deviation between initial and actual values. Additionally, the more efficient error correction for the duration of update procedure in the proposed filter. When the estimated error had emerged relatively stable, the estimation of proposed algorithm closer to the truth value.

Among the position, velocity and posture various state estimation aspects, the convergence velocity and accuracy had evident superiority than the existing EKF.

VII. CONCLUSIONS

Exercising simulation results on a right-invariant form of the extended Kalman filter, this article derived an motion state observer for double lower-limbs kinematics of pedestrian. With available IMU sensors on pedestrian's waist and lower limbs, the developed scheme has the capability to improve the accuracy and reliability of state estimation of pedestrian. We also included IMU biases in the state estimator and showed that, the proposed system has better performance than that of a commonly used EKF. Future work includes integration of the state estimation developed in this work to operate and even more complicated action in experiment, constructing a outdoor system.

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